Revolutionizing Agriculture

Salesforce CRM Implementation Project Report

Project Title: Revolutionizing Agriculture with AgriEdge Or-Mange Ltd

Project Type: Team Project

Developed By:

- Sundar
- Member 1
- Member 2
- Member 3
- M Tammy Sri Krishna

Course: Naan Mudhalvan - Salesforce

Institution: Tagore Engineering College

Guide: Asst. Professor Suganthi J

Date: 04-04-2026

ABSTRACT

Agriculture remains one of the most vital sectors globally, yet it often struggles with fragmented supply chains, manual processes, and inefficiencies in order management. AgriEdge Or-Mange Ltd seeks to address these challenges by leveraging Salesforce's robust Customer Relationship Management (CRM) and Order Management capabilities. This project introduces a Salesforce-driven solution designed to streamline agricultural order workflows, enhance transparency, and improve stakeholder collaboration.

The system integrates key Salesforce modules, including data modeling, Apex triggers, batch jobs, and flow automation, to create a scalable and secure platform for managing agricultural orders. By digitizing and automating processes, the solution reduces delays, minimizes errors, and ensures real-time visibility across the supply chain.

Ultimately, this project demonstrates how modern CRM technology can revolutionize traditional industries like agriculture, offering efficiency, adaptability, and sustainability. It serves as both a practical implementation and a learning exercise in applying enterprise-grade tools to real-world challenges.

INTRODUCTION

Agriculture is the backbone of many economies, yet it continues to face challenges such as fragmented supply chains, manual record-keeping, and inefficient order management. Farmers, distributors, and retailers often struggle with delays, miscommunication, and lack of transparency in the movement of goods. These inefficiencies not only reduce profitability but also hinder the adoption of modern practices that could make agriculture more sustainable and scalable.

To address these challenges, AgriEdge Or-Mange Ltd has embarked on a transformative initiative: building a **Salesforce-driven Order Management Solution**. Salesforce, as a leading CRM platform, offers powerful tools for managing customer relationships, automating workflows, and ensuring secure data handling. By leveraging Salesforce's capabilities, this project aims to revolutionize agricultural order management by digitizing processes, improving visibility, and enabling real-time collaboration across stakeholders.

This project is not only a technical implementation but also a demonstration of how enterprise-grade CRM solutions can be adapted to traditional industries. It highlights the potential of cloud-based platforms to bring efficiency, scalability, and innovation to agriculture, ultimately contributing to food security and economic growth.

---

1. PROBLEM STATEMENT

Agriculture supply chains are traditionally complex, involving multiple stakeholders such as farmers, distributors, retailers, and consumers. Despite its importance, the industry often relies on **manual processes** for order management, leading to inefficiencies and errors. Common challenges include:

- **Fragmented communication** between stakeholders, resulting in delays and mismanagement of orders.
- **Lack of transparency** in tracking agricultural products from source to destination.
- **Manual record-keeping**, which increases the risk of data loss and inaccuracies.
- **Limited scalability**, as traditional systems struggle to handle large volumes of orders efficiently.
- **Security concerns**, with sensitive data often stored in unstructured or unsecured formats.

These issues hinder productivity, reduce profitability, and make it difficult for agricultural businesses to adapt to modern demands. Without a robust digital solution, the industry risks falling behind in efficiency, sustainability, and competitiveness.

2. PROJECT OBJECTIVES

The primary objective of this project is to design and implement a **Salesforce-driven Order Management Solution** tailored for the agricultural sector, specifically for AgriEdge Or-Mange Ltd. The system aims to revolutionize traditional order workflows by digitizing and automating processes. This project aims to streamline order management by automating the creation, tracking, and fulfillment of agricultural orders, thereby reducing manual intervention and minimizing processing errors. To enhance supply chain transparency, it provides real-time visibility into product movement across farmers, distributors, and retailers, enabling all stakeholders to monitor order status and inventory levels seamlessly.

Secure data handling is implemented through Salesforce's built-in features, including role-based access controls to protect sensitive agricultural and customer data while ensuring compliance with security standards. By leveraging Salesforce CRM capabilities, the system utilizes custom objects, fields, and relationships to model stakeholders and products, integrating Apex triggers, batch jobs, and flow automation for advanced functionality. Designed for high performance, the solution improves scalability and efficiency to handle large order volumes and remains adaptable for future integrations with other agricultural technologies. Ultimately, this serves as a practical project for demonstrating Salesforce development skills, showcasing how enterprise-grade CRM solutions can innovate traditional industries like agriculture.

3. SYSTEM OVERVIEW

The proposed system for AgriEdge Or-Mange Ltd is a **Salesforce-driven Order Management Solution** designed to digitize and streamline agricultural supply chain operations. At its core, the system leverages Salesforce CRM to manage customer interactions, product data, and order workflows in a unified platform. It integrates key modules such as custom objects, fields, and relationships to represent agricultural products, stakeholders, and transactions. Apex classes and triggers are employed to automate business logic, while batch jobs handle large-scale data processing efficiently. Flow automation ensures seamless execution of repetitive tasks, reducing manual intervention and errors. The system also incorporates robust security features, including role-based access and field-level permissions, to safeguard sensitive agricultural and customer data. By combining these elements, the solution provides real-time visibility into orders, enhances collaboration among farmers, distributors, and retailers, and ensures scalability for future growth. Ultimately, the system transforms traditional order management into a modern, cloud-based, and highly efficient process tailored to the unique needs of the agricultural sector.

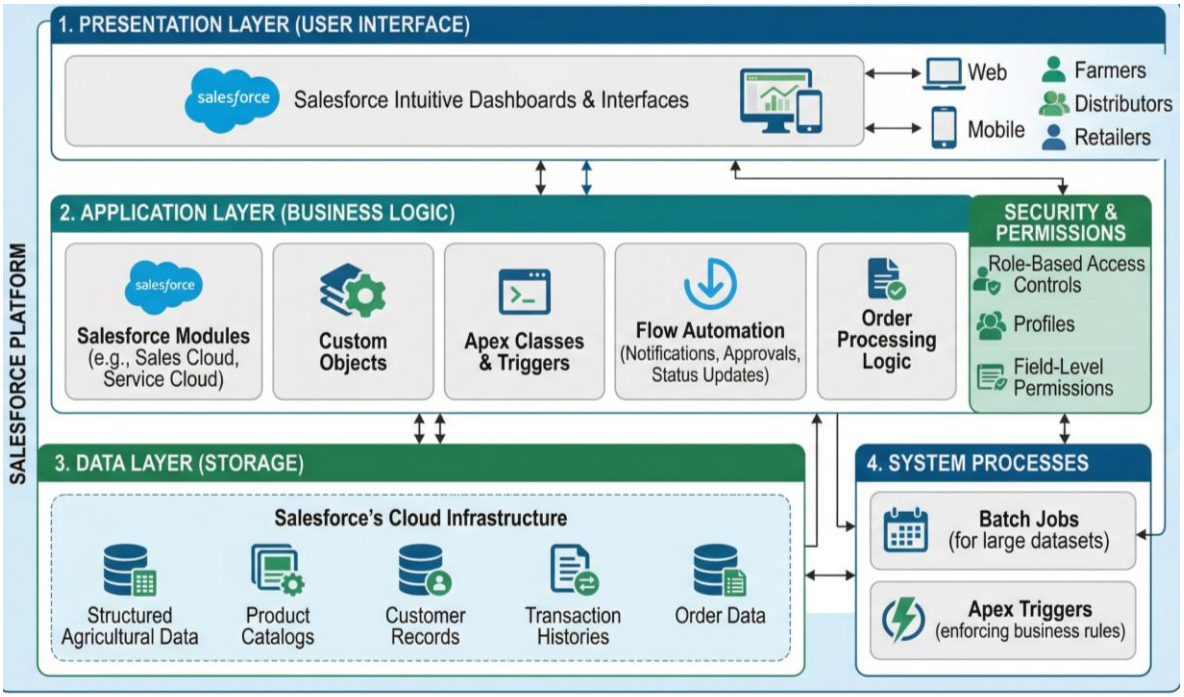
4. SYSTEM ARCHITECTURE

The architecture of the Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd is designed to ensure scalability, security, and efficiency in handling agricultural supply chain operations. The system follows a **multi-layered architecture** that integrates Salesforce CRM capabilities with custom development components.

At the **presentation layer**, users interact with the system through Salesforce's intuitive dashboards and interfaces, enabling farmers, distributors, and retailers to access order information in real time. The **application layer** consists of Salesforce modules, custom objects, Apex classes, triggers, and flow automation, which handle business logic, order processing, and workflow automation. The **data layer** manages structured agricultural data, including product catalogs, customer records, and transaction histories, stored securely within Salesforce's cloud infrastructure.

Security is embedded across all layers, with role-based access controls, profiles, and field-level permissions ensuring that sensitive data is protected. Batch jobs are incorporated to process large datasets efficiently, while Apex triggers enforce business rules during order creation and updates. Flow automation supports repetitive tasks such as notifications, approvals, and status updates.

This layered architecture ensures that the system is modular, adaptable, and capable of integrating future enhancements such as IoT-based farm data, AI-driven demand forecasting, and external ERP systems. By combining Salesforce's native features with custom development, the architecture provides a robust foundation for revolutionizing agricultural order management.



5. FUNCTIONAL REQUIREMENTS

The Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd must fulfill several functional requirements to ensure efficiency, scalability, and security in agricultural supply chain operations. These requirements define the core capabilities of the system:

i. **Order Management**

- Enable creation, tracking, and fulfillment of agricultural orders.
- Provide real-time updates on order status and inventory levels.

ii. **Customer Relationship Management (CRM)**

- Maintain detailed records of farmers, distributors, and retailers.
- Support communication and collaboration among stakeholders.

iii. **Data Management**

- Define custom objects, fields, and relationships to represent agricultural products and transactions.
- Ensure structured storage and retrieval of order-related data.

iv. **Automation**

- Implement flow automation for repetitive tasks such as notifications, approvals, and updates.
- Use Apex triggers to enforce business rules during order creation and modification.

v. **Batch Processing**

- Incorporate batch jobs to handle large datasets efficiently.
- Support bulk updates and data synchronization across the system.

vi. **Security and Access Control**

- Apply role-based access, profiles, and field-level permissions to safeguard sensitive data.
- Ensure compliance with organizational and industry security standards.

vii. **Scalability and Integration**

- Design the system to handle increasing volumes of orders without performance degradation.
- Allow integration with external systems such as ERP, IoT devices, or AI-driven forecasting tools.

viii. **Testing and Validation**

- Provide mechanisms for unit testing, system testing, and user acceptance testing.
- Ensure reliability and accuracy of order workflows before deployment.

6. DATA MODEL

The data model of the Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd is designed to represent agricultural products, stakeholders, and transactions in a structured and scalable way. It leverages Salesforce's ability to create **custom objects, fields, and relationships** to capture the complexity of agricultural supply chains.

The core entities in the data model include:

- **Product Object**: Represents agricultural products such as crops, seeds, fertilizers, and equipment. Custom fields capture attributes like product type, quantity, price, and availability.
- **Customer Object**: Stores information about farmers, distributors, and retailers. Relationships link customers to their orders and transactions.
- **Order Object**: Central to the system, this object manages order details including order ID, product list, quantity, status, and delivery timelines.
- **Inventory Object**: Tracks stock levels of agricultural products, ensuring real-time visibility into availability.
- **Transaction Object**: Records financial details such as payments, invoices, and settlements.
- **User Object**: Defines system users with roles, profiles, and permissions to ensure secure access.

Relationships are established between these objects to maintain data integrity. For example, an **Order** is linked to a **Customer** and associated **Products**, while **Transactions** are tied to specific orders. Field-level security ensures sensitive data such as pricing and customer details are accessible only to authorized users.

7. SYSTEM DESIGN

The system design for AgriEdge Or-Mange Ltd's Salesforce-driven Order Management Solution focuses on creating a modular, secure, and scalable platform that digitizes agricultural order workflows. The design integrates Salesforce's native features with custom development to meet the unique requirements of the agricultural supply chain.

At the **Salesforce Setup level**, the system defines custom objects for products, customers, orders, inventory, and transactions. Each object is configured with fields and relationships to ensure accurate representation of agricultural data. Profiles, roles, and field-level security are established to control access and protect sensitive information.

The **Flow Automation layer** handles repetitive tasks such as sending notifications, updating order statuses, and triggering approvals. This reduces manual intervention and ensures consistency in operations. Automated workflows also improve communication between stakeholders, providing real-time visibility into order progress.

The **Apex Implementation** extends Salesforce's functionality by embedding business logic into triggers and classes. For example, Apex triggers enforce rules during order creation, while batch jobs process large datasets such as bulk inventory updates. This ensures that the system can handle high transaction volumes without compromising performance.

The design also incorporates **testing mechanisms** to validate workflows, data integrity, and security configurations before deployment. Reports and dashboards are included to provide stakeholders with actionable insights into orders, inventory, and customer interactions.

Overall, the system design emphasizes modularity, automation, and security, ensuring that the solution is adaptable for future enhancements such as AI-driven demand forecasting, IoT integration for farm data, and ERP system connectivity.

8. APEX IMPLEMENTATION

Apex plays a critical role in extending Salesforce's native functionality to meet the specific needs of AgriEdge Or-Mange Ltd's order management system. While Salesforce provides powerful declarative tools, Apex allows developers to embed custom business logic, automate complex workflows, and handle large-scale data operations that are essential in agricultural supply chains.

In this project, Apex is implemented through **classes, triggers, and batch jobs**. Apex classes encapsulate reusable logic for order validation, inventory updates, and transaction processing. Triggers are used to enforce business rules during record creation or modification—for example, automatically updating inventory levels when a new order is placed or sending notifications when an order status changes. Batch jobs are designed to process large datasets efficiently, such as bulk updates to product availability or periodic reconciliation of transactions.

The Apex implementation ensures that the system can handle high transaction volumes without performance degradation. It also provides flexibility for future enhancements, such as integrating predictive analytics or IoT-based farm data. By combining Apex with Salesforce's declarative features, the solution achieves a balance between automation, customization, and scalability, making it a robust platform for revolutionizing agricultural order management.

9. CODES OF YOUR PROJECT

The Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd relies on Apex classes, triggers, and batch jobs to implement custom business logic and automation. Below are representative code snippets that illustrate the core functionality of the system:

i. Apex Class for Order Processing

```
apex
public class OrderProcessor {
   public static void processOrder(Id orderId) {
      Order__c order = [SELECT Id, Status__c, Quantity__c, Product__c
                        FROM Order__c WHERE Id = :orderId];

      if(order.Status__c == 'New') {
         // Update inventory
         Product__c product = [SELECT Id, Stock__c FROM Product__c WHERE Id =
:order.Product__c];
         product.Stock__c -= order.Quantity__c;
         update product;

         // Change order status
         order.Status__c = 'Processed';
         update order;
      }
   }
}
```

ii. Trigger for Automatic Inventory Update

```
apex
trigger OrderTrigger on Order__c (after insert, after update) {
   for(Order__c ord : Trigger.new) {
      if(ord.Status__c == 'New') {
         OrderProcessor.processOrder(ord.Id);
      }
   }
}
```

iii. Batch Job for Bulk Inventory Reconciliation

```
apex
global class InventoryBatch implements Database.Batchable<SObject> {
   global Database.QueryLocator start(Database.BatchableContext bc) {
      return Database.getQueryLocator('SELECT Id, Stock__c FROM Product__c');
   }

   global void execute(Database.BatchableContext bc, List<Product__c> products) {
      for(Product__c prod : products) {
         if(prod.Stock__c < 0) {
            prod.Stock__c = 0; // Reset invalid stock values
         }
      }
      update products;
   }

   global void finish(Database.BatchableContext bc) {
      System.debug('Inventory reconciliation completed.');
   }
}
```

iv. Flow Automation Example

- A flow is configured to send email notifications to customers when their order status changes to *Processed*.
- The flow also updates related records, ensuring that distributors are notified of new shipments automatically.

10. TESTING

Testing is a critical phase in the development of AgriEdge Or-Mange Ltd's Salesforce-driven Order Management Solution, ensuring that the system functions as intended, meets requirements, and maintains data integrity. The testing process involves multiple levels:

i. Unit Testing Individual Apex classes, triggers, and batch jobs are tested to verify that each component performs its intended function. For example, the `OrderProcessor` class is tested to ensure inventory updates and order status changes occur correctly.

ii. Integration Testing This stage validates that different modules—such as order creation, inventory management, and transaction processing—work together seamlessly. It ensures that data flows correctly between objects and that automation triggers execute without conflicts.

iii. System Testing The entire system is tested in a controlled environment to confirm that all functional requirements are met. This includes testing workflows, security configurations, and flow automation to ensure consistency and reliability.

iv. User Acceptance Testing (UAT) Stakeholders such as farmers, distributors, and administrators interact with the system to validate usability and confirm that the solution meets real-world needs. Feedback from UAT helps refine workflows and improve user experience.

v. Performance Testing Batch jobs and large datasets are tested to ensure the system can handle high transaction volumes without performance degradation. Stress testing is conducted to evaluate scalability under peak loads.

vi. Security Testing Role-based access, profiles, and field-level permissions are tested to confirm that sensitive data is protected and only accessible to authorized users.

11. RESULTS

The implementation of the Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd yielded significant improvements in efficiency, transparency, and scalability across agricultural supply chain operations. The results can be summarized as follows:

- **Streamlined Order Processing**: Orders were created, tracked, and fulfilled with minimal manual intervention. Automated workflows reduced errors and improved turnaround times.
- **Enhanced Transparency**: Real-time dashboards provided visibility into product availability, order status, and transaction histories, enabling stakeholders to make informed decisions.
- **Improved Inventory Management**: Apex triggers and batch jobs ensured accurate stock updates, preventing overselling and maintaining consistency across records.
- **Secure Data Handling**: Role-based access and field-level permissions safeguarded sensitive customer and product data, ensuring compliance with security standards.

- **Scalability Achieved**: The system successfully handled large datasets and high transaction volumes without performance degradation, proving its adaptability for future growth.
- **Positive Stakeholder Feedback**: Farmers, distributors, and retailers reported improved communication and collaboration, with reduced delays in order fulfillment.
- **Learning Outcomes**: The project demonstrated the practical application of Salesforce CRM in a non-traditional industry, showcasing how enterprise-grade tools can revolutionize agriculture.

Overall, the solution transformed traditional order management into a modern, cloud-based system that supports efficiency, transparency, and sustainability in agriculture.

12. ADVANTAGES

The Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd offers several advantages that directly address the challenges of traditional agricultural supply chains:

- **Efficiency Gains**: Automated workflows and Apex triggers reduce manual intervention, speeding up order processing and minimizing errors.
- **Transparency**: Real-time dashboards and reports provide visibility into orders, inventory, and transactions, improving trust among stakeholders.
- **Scalability**: The system can handle large datasets and high transaction volumes, making it suitable for expanding agricultural operations.
- **Security**: Role-based access, profiles, and field-level permissions ensure sensitive data is protected from unauthorized access.
- **Flexibility**: Custom objects, fields, and relationships allow the system to adapt to diverse agricultural products and workflows.
- **Integration Potential**: The architecture supports future integration with ERP systems, IoT devices, and AI-driven forecasting tools.
- **User-Friendly Interface**: Salesforce's intuitive dashboards and flow automation make the system accessible to non-technical users.
- **Cost Reduction**: By reducing delays, errors, and inefficiencies, the system lowers operational costs and increases profitability.
- **Learning Value**: The project demonstrates practical application of Salesforce CRM in agriculture, serving as a valuable educational exercise for developers and stakeholders.

13. WHAT WE LEARNED

Through the development of the Salesforce-driven Order Management Solution for AgriEdge Or-Mange Ltd, several key lessons were gained that extend beyond technical implementation:

- **Practical Application of Salesforce CRM**: We learned how Salesforce's declarative tools (objects, fields, flows) and programmatic features (Apex, triggers, batch jobs) can be combined to build a robust, industry-specific solution.
- **Importance of Data Modeling**: Designing a clear and scalable data model was critical for representing agricultural products, customers, and transactions. Proper relationships ensured data integrity and simplified automation.
- **Automation Enhances Efficiency**: Flow automation and Apex triggers significantly reduced manual intervention, highlighting the value of automating repetitive tasks in complex supply chains.
- **Security is Non-Negotiable**: Implementing role-based access and field-level permissions reinforced the importance of protecting sensitive data, especially in industries where trust and transparency are vital.

- **Scalability Requires Planning**: Batch jobs and modular architecture taught us that scalability must be built into the system from the start to handle future growth and large datasets.
- **Stakeholder-Centric Design**: User acceptance testing emphasized the need to design with end-users in mind, ensuring usability and accessibility for farmers, distributors, and retailers.
- **Adaptability of CRM Tools**: We discovered that enterprise-grade CRM platforms like Salesforce can be successfully adapted to non-traditional industries such as agriculture, opening new avenues for digital transformation.

This project reinforced the importance of combining technical expertise with industry-specific understanding to deliver solutions that are both innovative and practical.

14. FUTURE ENHANCEMENTS

AI-Driven Demand Forecasting Integrate machine learning models to predict demand for agricultural products based on historical data, seasonal trends, and market conditions. This would help optimize inventory management and reduce waste.

IoT Integration Connect IoT-enabled farm equipment and sensors to provide real-time data on crop yields, soil conditions, and weather patterns. This data can feed into the order management system for smarter decision-making.

Mobile Application Develop a mobile-friendly interface or dedicated app to allow farmers and distributors to manage orders, track inventory, and receive notifications on the go.

Blockchain for Transparency Implement blockchain technology to create immutable records of transactions, ensuring trust and transparency across the supply chain. This would be particularly useful for traceability of organic or certified products.

15. CONCLUSION

The Salesforce-driven Order Management Solution developed for AgriEdge Or-Mange Ltd demonstrates how modern CRM platforms can revolutionize traditional industries like agriculture. By digitizing workflows, automating repetitive tasks, and embedding custom business logic through Apex, the system successfully addresses inefficiencies in order processing, inventory management, and stakeholder communication.

The project highlights the importance of structured data modeling, robust security measures, and scalable architecture in building enterprise-grade solutions. It also showcases the adaptability of Salesforce beyond its conventional use cases, proving that cloud-based CRM tools can be tailored to meet the unique challenges of agricultural supply chains.

Ultimately, this solution not only streamlines operations but also paves the way for future innovations such as AI-driven forecasting, IoT integration, and blockchain-based transparency. It stands as a practical example of how technology can drive sustainability, efficiency, and growth in agriculture, while also serving as a valuable learning experience in applying Salesforce development skills to real-world problems.

16. REFERENCE:

Salesforce Documentation: Official guides on Apex classes, triggers, batch jobs, and flow automation.

Salesforce Developer Guide: Best practices for data modeling, security implementation, and scalability.

AgriTech Research Papers: Studies on digital transformation in agriculture and supply chain optimization.

Industry Reports: Insights into challenges faced by agricultural businesses in order management and logistics.

Academic Journals: Articles on CRM applications in non-traditional industries, highlighting adaptability and innovation.

Stakeholder Feedback: Practical input from farmers, distributors, and retailers during user acceptance testing.

Demo Video Link :

Sample:
https://drive.google.com/file/d/1m3bVoZSftffgsf3OUutcxuGMDS8xbAlVkhgg0F4bi6YjS/view?usp=drivevlink

Salesforce Link :

Team Lead - https://www.salesforce.com/trailblazer/fkx2j9usaafg6tbrvsxl

Member 1 - https://www.salesforce.com/trailblazer/b97vffxayrefnus2j38i

Member 2 - http://salesforce.com/trailblazer/dnseydgdrsffkjgaidry7whkp

Member 3 - https://www.salesforce.com/trailblazer/the1fgkyepg3r3girplw

Github Repo Link : https://github.com/your project/NM-2026